

MAYO CLINIC ANALYTICS DASHBOARD

COMP50063 - DATA ANALYTICS

CB NUMBER - CB015744

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Problem & Context

- Healthcare organizations generate a massive amount of data.
- Each hospital branch often stores data in a separate system.
- There is no single dashboard to monitor all hospital branches.
- There is no effective competitor benchmarking mechanism.
- Data quality issues will be reduced by the accuracy of analysis and reporting.
- Real-time KPI dashboards support executives in making faster decisions.

(Statista, 2024)

Problem & Context

Project Aim

- Designing a 3-page interactive Power BI dashboard for Mayo Clinic.
- The Mayo Clinic dashboard should contain comparisons of branch performance, benchmark competitors, and support data-driven decisions.

	ABC	Doctor_Notes	ABC	Admit_Date	ABC	Discharge_Date	ABC	Bill_Amount	ABC
	100%	Valid	100%	Valid	98%	Valid	100%	Valid	94%
	0%	Error	0%	Error	0%	Error	0%	Error	0%
	0%	Empty	0%	Empty	2%	Empty	0%	Empty	6%
1		Disease-modifying therapy initiated.		9/6/2023		9/21/2023		19132.09	J
2		Discussed potential for deep brain stimulation surgery.		7/12/2023		7/20/2023		null	M
3	s	Patient advised to monitor blood sugar levels regularly.		1/17/2022		1/27/2022		null	C
4		Referred to physical therapy to improve mobility and balance.		3/11/2023		3/31/2023		9994.45	M
5		Started on antibiotics for bacterial infection.		7/28/2022		8/26/2022		2901.89	C

Traditional Vs Big Data vs IoT

1

Traditional Processing

Tools: Excel, Paper Reports, Manual Entry

Speed: Batch processing in weekly/monthly reports.

Scale: Handling a maximum of a thousand rows.

Flexibility: Static, non-interactive reports.

Cost: Low cost but a high-time-consuming process.

Example: Spreadsheet of monthly patient count.

2

Big Data Approach

Tools: Power BI,

Speed: Near real-time processing

Scale: Handling billions of rows.

Flexibility: Dynamic, Interactive dashboard

Cost: Higher but faster insights

Example: Live performance of the hospital dashboard.

3

IOT in Healthcare

Smart beds: Monitoring patients' movements and preventing falls

Wearable monitors: Providing real-time heart data.

Automated safety alerts for medication.

RFID Tracking: Instantly locating medical equipment.

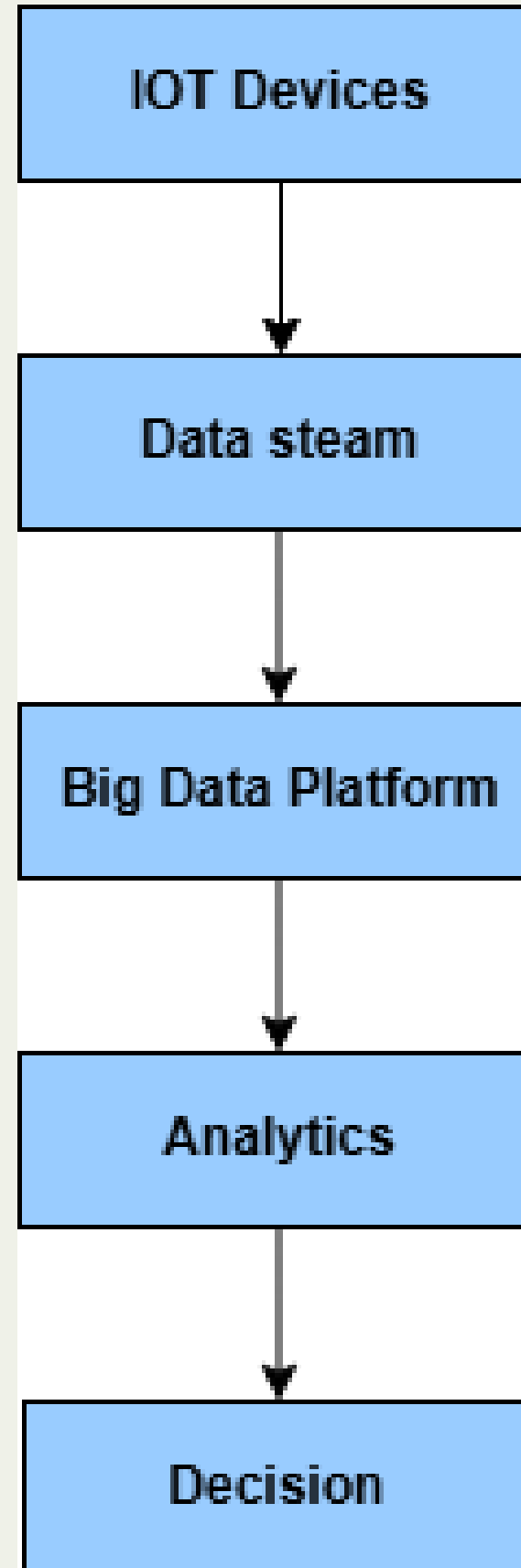
Predictive maintenance:

Detecting medical equipment issues before failures.

Patient flow sensors: Improving waiting room management.

(IBM, 2023)

IOT Data Flow



Why Big Data Over Traditional?

- Continuous IoT data streams cannot be handled by Traditional tools.
- Only real-time data creates real-time data.
- The volume of healthcare data exceeds excel capacity.
- Distributed computing simultenancy enables analysis across multiple hospitals, departments and data sources.



Dashboard selection & Justification

- Source – Kaggle [Hospital patient Records (Jan 2021 – July 2024)]
- Author – devildyno (2024)
- Original rows – 1000 patient records
- Final rows after cleaning – 982
- Columns – 15 (12 Original and 3 engineered)
- Hospitals – Mayo Clinic, Johns Hopkins, Cleveland Clinic
- Cities – New York, Chicago, Los Angeles

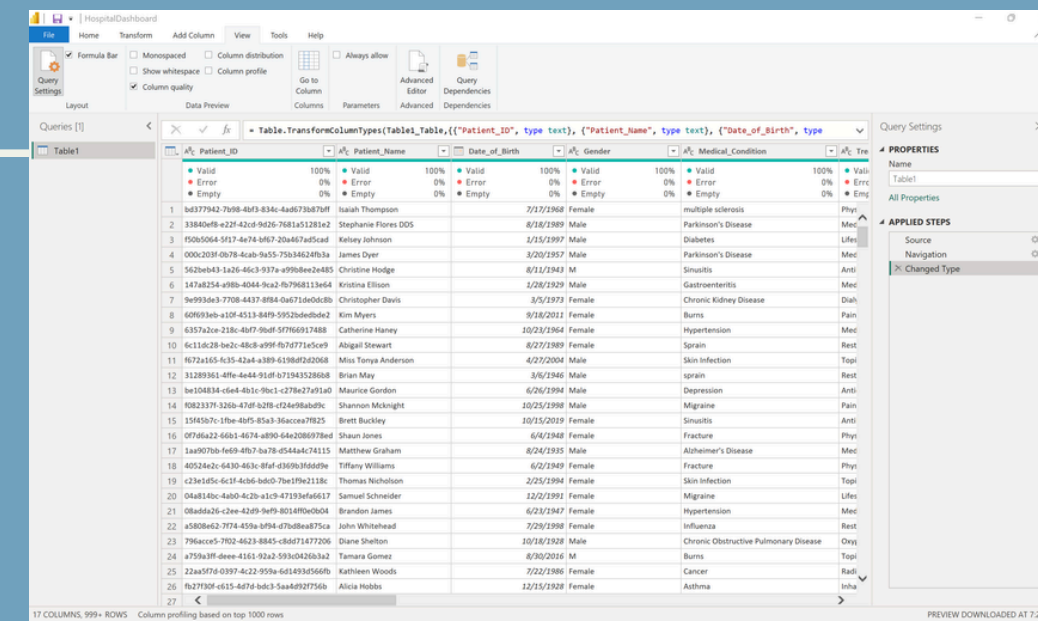


Table: TransformColumnTypes(Table1, [{"Patient_ID", type text}, {"Patient_Name", type text}, {"Date_of_Birth", type text}, {"Gender", type text}, {"Medical_Condition", type text}, {"", type text}, {"", type text}, {"", type text}])

Patient_ID	Patient_Name	Date_of_Birth	Gender	Medical_Condition			
16377942-7698-4853-834c-4a68736876f8	Isaiah Thompson	7/17/1968	Female	multiple sclerosis			
3384049b-e22f-42c0-9626-7681a51281a2	Stephanie Flores DDS	8/18/1989	Male	Parkinson's Disease			
75002064-9173-4a7e-0967-20a617a05a0d	Kelley Johnson	1/15/1997	Male	Diabetes			
0002305f-0b78-4a0b-9a53-753a6236783a	James Oyer	8/20/1957	Male	Parkinson's Disease			
562be043-1a26-46c3-937a-a996be214851	Christina Hodge	8/11/1943	M	Smualls			
1a7a8254-a98b-4044-9a2a-b796811364d	Kristina Ellison	1/28/1929	Male	Gastroenteritis			
9e9336a3-7708-4a37-894d-0a71d6a0d8b1	Christopher Davis	3/5/1973	Female	Chronic Kidney Disease			
60933ab0-a10f-4313-8a95-5952bde0b6d2	Kim Myers	9/18/2011	Female	Burns			
63532a2a-2186-407f-90af-9776d911488	Catherine Henry	10/22/1964	Female	Hypertension			
6c11a23b-ba2c-48c6-a99b-6c7677125a5d	Abigail Stewart	8/27/1989	Female	Sprain			
1672a165-fc35-42a4-a389-619baf220068	Miss Tanya Anderson	4/27/2004	Male	Skin Infection			
31289361-4f4e-4a44-918f-b71943528608	Brian May	3/6/1946	Male	sprain			
1e104834-c6a4-401c-99c1-c278a27a91a0	Maurice Gordon	6/26/1994	Male	Depression			
0982337f-320b-47df-b29b-c26a98a0d9c	Shannon McKnight	10/25/1998	Male	Migraine			
11463b7c-19ae-4d95-82a3-30aace78325	Brett Buckley	10/15/2019	Female	Smualls			
0766a622-6603-4678-6a90-64a2080878ad	Shaun Jones	6/4/1948	Female	Fracture			
1a9507db-f6d9-487f-ba78-d544a4c74115	Matthew Graham	8/24/1935	Male	Alzheimer's Disease			
40524a2c-6430-463c-8f4f-d360b36a0d9e	Tiffany Williams	6/2/1949	Female	Fracture			
c23a1d5c-6c1f-4c06-ba60-79a279e2118e	Thomas Nicholson	2/25/1994	Female	Skin Infection			
0a48140c-4a40-4c2b-a3c9-f7193fa661f7	Samuel Schneider	12/2/1991	Female	Migraine			
08a6a26c-c3ee-42d9-9a7b-801490a0604	Brandon James	6/2/1947	Female	Hypertension			
a5806a62-7074-455e-b954-c7bd8ea8751a	John Whitehead	7/29/1998	Female	Influenza			
796ace5c-7802-4623-8845-c8a071a77206	Diane Shelton	10/18/1928	Male	Chronic Obstructive Pulmonary Disease			
a795a3ff-deee-4165-92a2-593a0426b3a2	Tamara Gomez	8/30/2016	M	Burns			
22a45f7a-0397-4c22-959a-6d1893d36d8b	Kathleen Woods	7/22/1986	Female	Cancer			
1627930f-a515-4d7e-bd63-5aa89207756b	Alicia Hobbs	12/15/1928	Female	Asthma			

Column statistics	
Count	982
Error	0
Empty	0
Distinct	661
Unique	421
Min	7/12/2021
Max	7/11/2024
Average	1/10/2023

Why was this dataset selected?

- **Temporal Richness** – It covers 3.5 years of data & enabling trend analysis over time.
- **Healthcare Domain** – Representing the fastest-growing BI application area.
- **Right Scale** – Manageable for analysis & it's large enough to generate meaningful ETL insights.
- **ETA Suitability** – It contains realistic data quality issues.

[illegible]

Big Data Fit - The 5 Vs

Volume – The dataset is relatively small, but the proposed architecture can scale to handle millions of rows.

Velocity – Time-based healthcare data is supporting real-time monitoring and analysis.

Variety – This dataset includes data types such as numerical, categorical, and text.

Veracity – Identified 7 data quality issues and resolved them through the ELT process.

Value – The analysis produced 4 actionable recommendations.

(BEYER AND LANEY, 2012)

ELT PROCESS

Issue	Records Affected	Solution
Duplicate Records	20	Removed using Remove Duplicates
Inconsistent gender values (M/F vs Male/Female)	38	Standardized using Replace Values
Inconsistent medical condition formatting	47	Applied Capitalized Each Word.
Invalid bill amount entries ("ERROR")	29	Replaced with null and converted to decimal formal.
Missing bill amount values	57	Replaced with median value of \$3,227.47
Inconsistent department (UPPER CASE)	33	Applied Capitalize Each Word
Missing admit data records	18	Removed incomplete rows

Total data quality issues resolved: 242

Query Settings

PROPERTIES

Name

Table1

All Properties

APPLIED STEPS

Source

Navigation

Changed Type

Removed Duplicates

Replaced Value

Replaced Value1

Capitalized Each Word

Replaced Value2

Changed Type1

Replaced Value3

Capitalized Each Word1

Removed Blank Rows

Filtered Rows

✕ Changed Type2

CRITICAL ELT DECISIONS

Why use the median instead of the Mean?

- Mean Bill amount is \$9,062
- Median Bill amount is \$3,227.47
- Healthcare billing contains high-value cases.
- For missing data, the median provides a more representative value.
- The mean can give misleading KPI s.

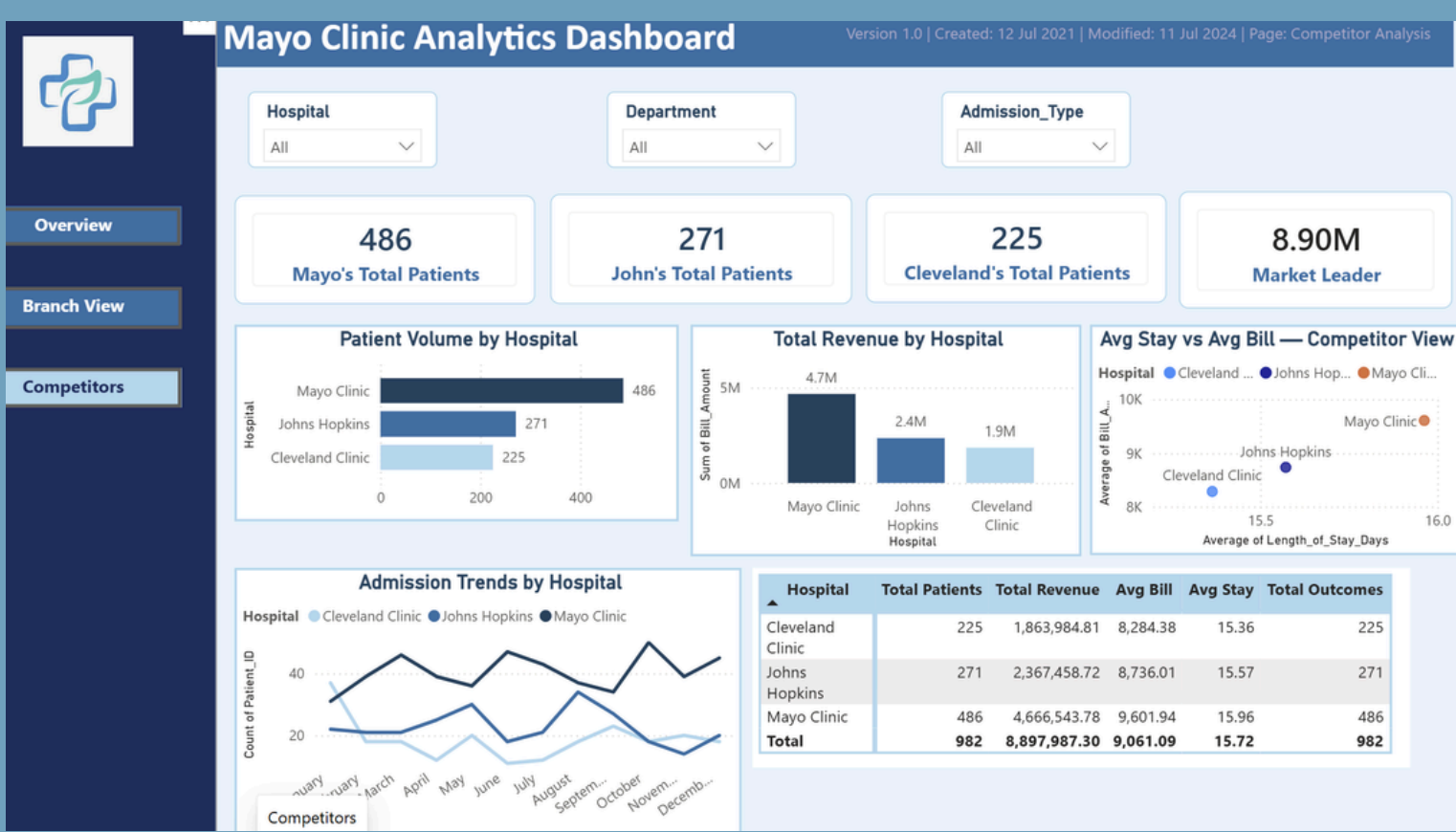
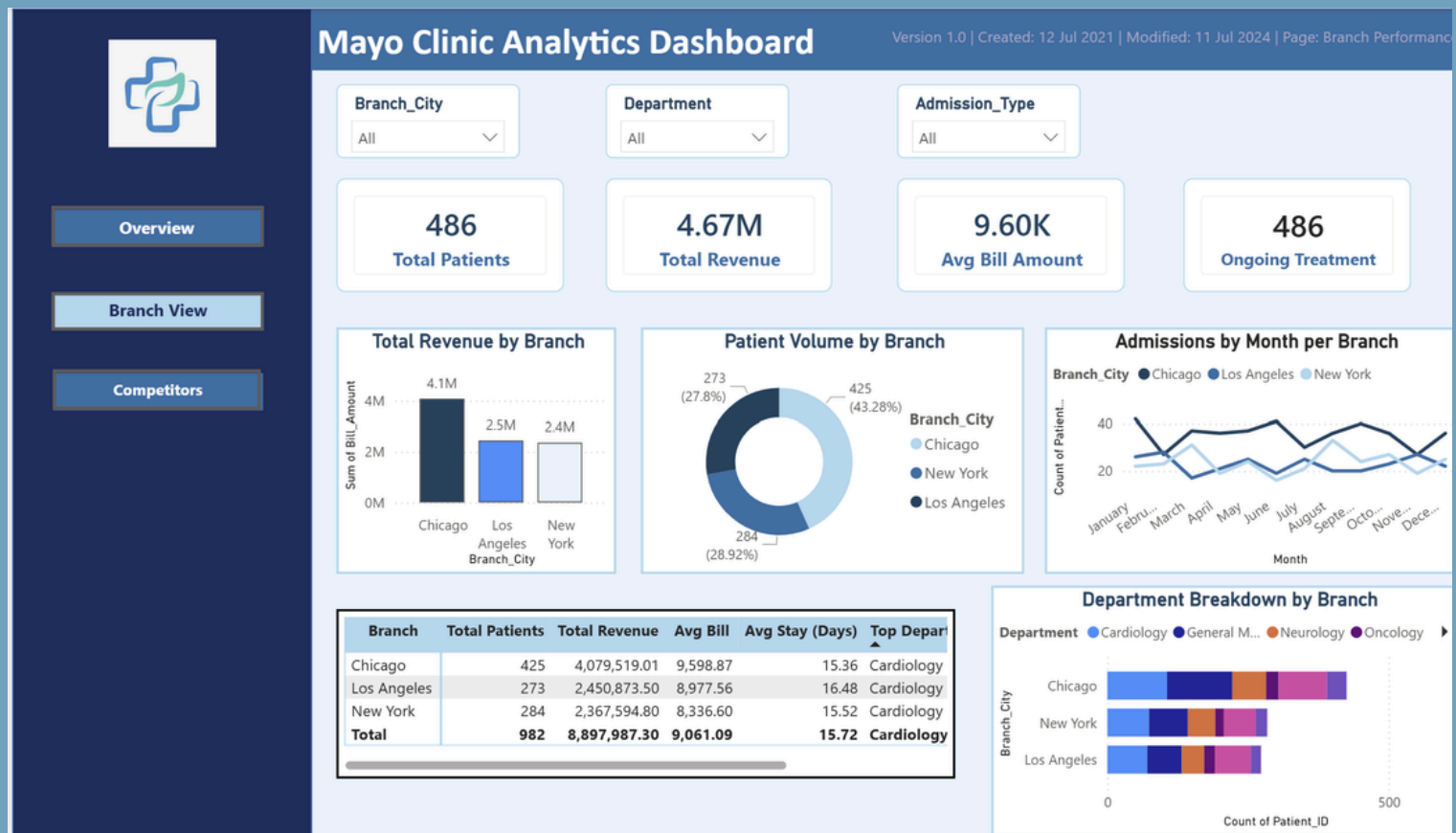
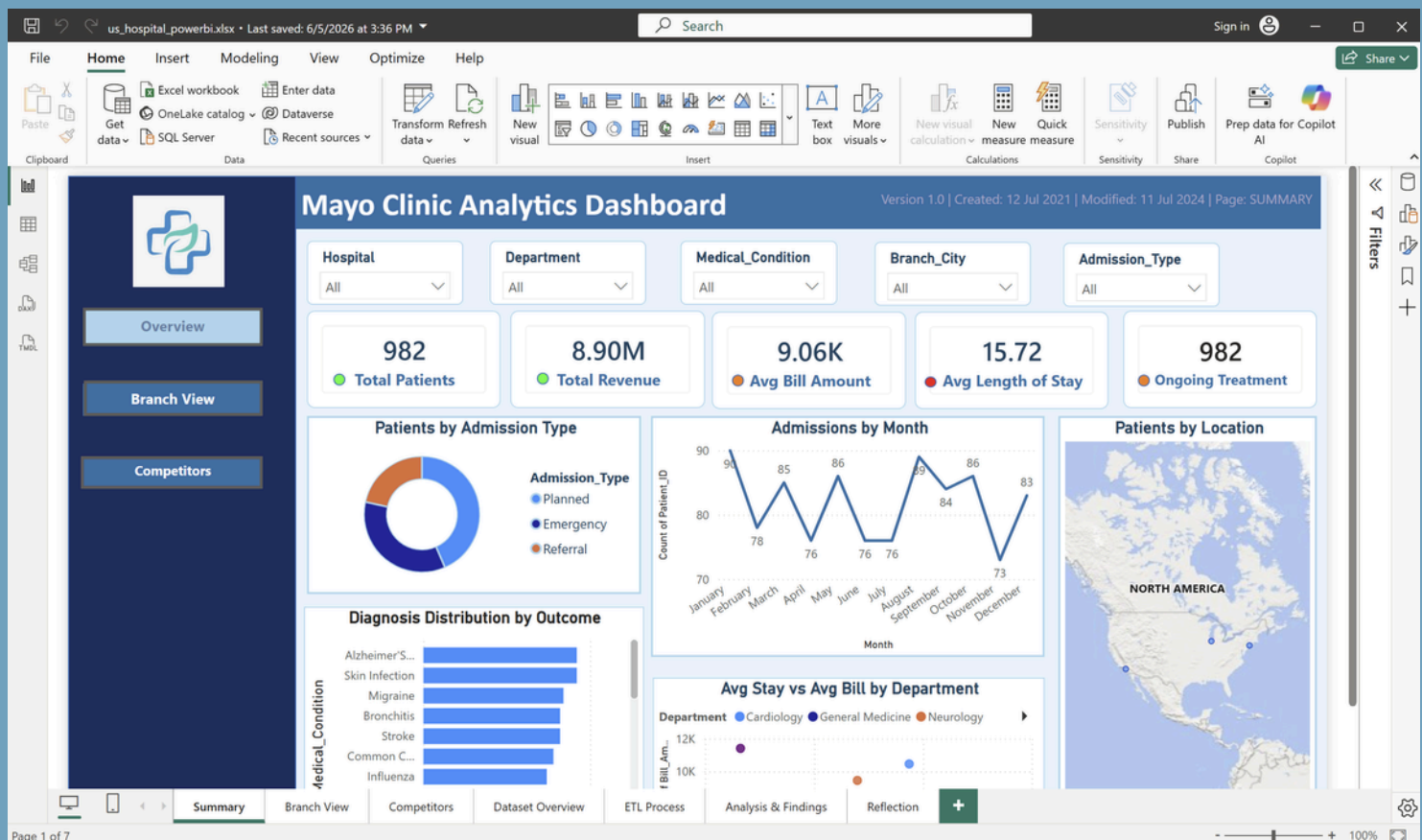
Queries [1]	<	  	= List.Median(#"Changed Type1"[Bill_Amount])
1 ² ₃ Table1			3227.475

CRITICAL ELT DECISIONS

Why remove missing admit dates better than of imputing?

- These admission dates are important for time-based analysis and trend reporting.
- These imputing dates can create artificial trends as well as misleading results.
- By removing these records, maintain data accuracy.

DASHBOARD DEMO



KEY FINDINGS

- Chicago is the top-performing branch of Mayo Clinic
It generates \$4.1 million in revenue.
- Mayo clinic 49.3% market share, John Hopkins, 28.6%, and the Cleveland clinic 22.1 % leads the market.
- The average patient length of stay is 15.72 days, which is 31%.
- The Cardiology department is the most in-demand across all branches and competitor hospitals.

Branch	Total Patients	Total Revenue	Avg Bill	Avg Stay (Days)	Top Depart
Chicago	425	4,079,519.01	9,598.87	15.36	Cardiology
Los Angeles	273	2,450,873.50	8,977.56	16.48	Cardiology
New York	284	2,367,594.80	8,336.60	15.52	Cardiology
Total	982	8,897,987.30	9,061.09	15.72	Cardiology

Recommendations

- Investing in the Chicago hospital → It generates the highest revenue and patient volume.
- Reducing average length of stay → It is above industry benchmark
- Strengthening New York operations → It is the weakest performing branch located in the healthcare market.
- Monitoring Johns Hopkins in Chicago

BUSINESS RECOMMENDATIONS

1. EXPAND CHICAGO OPERATIONS

Chicago generates 43% of revenue with highest patient volume.
Investment in capacity expansion would yield highest ROI.

2. REDUCE LENGTH OF STAY

Average stay of 15.72 days exceeds industry benchmark by 31%.
Process optimization in General Medicine could reduce costs significantly.

3. STRENGTHEN NEW YORK PRESENCE

New York generates lowest revenue (\$2.4M) despite being largest US market.
Targeted marketing and specialist recruitment recommended.

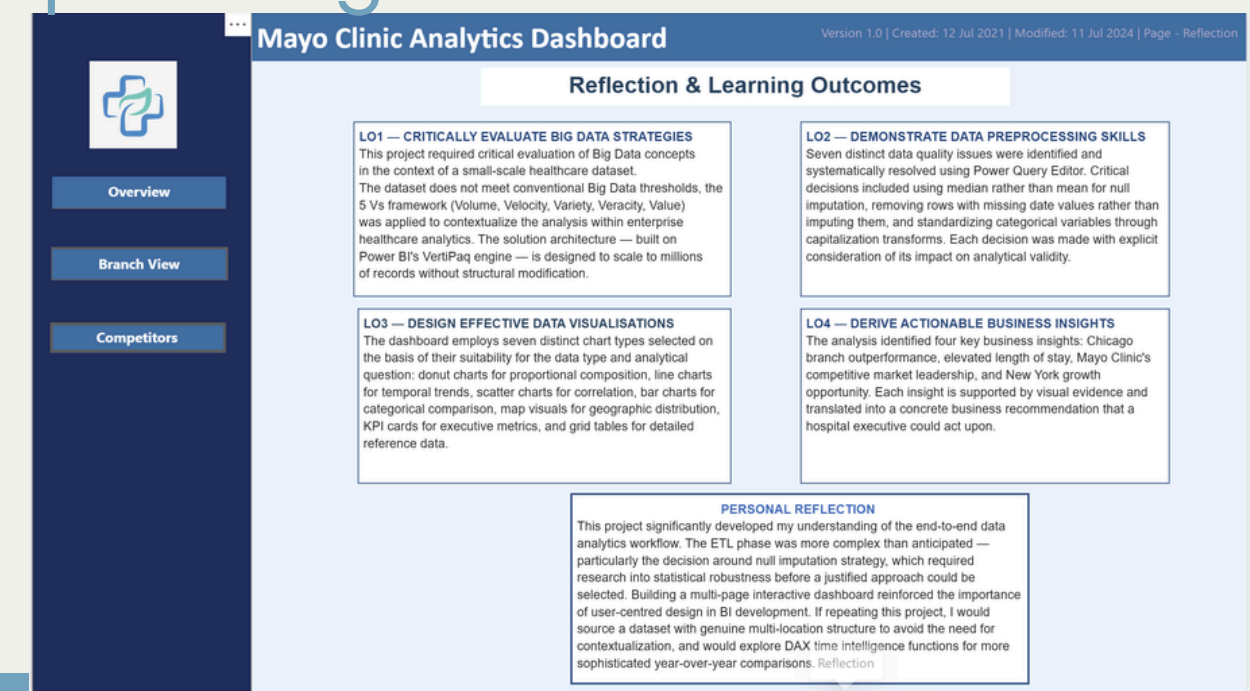
4. MONITOR COMPETITOR GROWTH

Johns Hopkins showing strong performance in Chicago — same city as Mayo Clinic's top branch.
Competitive strategy review recommended.

Challenges & Reflection

Challenges that I actually faced

- Fixing the KPI traffic light indicators → Card & a coloured circle were implemented
- Contextualizing the database → The healthcare scenario adopted around available data.
- Managing dashboard space → Navigation planning has been done



Challenges & Reflection

Limitations

- Using a simulated data set → These findings cannot directly apply to the real Mayo Clinic.
- Using synthetic competitor data → It doesn't represent actual market performance.
- Using a small-scale data set → The dataset doesn't require an enterprise big data platform.

Learning Outcomes

LO1 – Critically evaluating big data strategies

- Applied 5v framework
- While demonstrating scalability, the dataset limitations were acknowledged.

LO2 – Presenting logical arguments for the choice of approaches.

- It has justified data cleaning methods and visualization decisions.
- It has used best practice design principles.

LO3– Researching, Evaluating, and Analyzing Big Data Practices

- It has applied big data frameworks.
- It is an evaluated Power BI solution.

References

- Beyer, M. and Laney, D. (2012). The Importance of 'Big Data': A Definition. Gartner Research. Available at: <https://www.gartner.com> [Accessed: 3 June 2026].
- IBM (2023). What is Big Data? IBM Think. Available at: <https://www.ibm.com/think/topics/big-data> [Accessed: 3 June 2026].
- Statista (2024). Volume of data generated in healthcare in the United States. Statista Research Department. Available at: <https://www.statista.com> [Accessed: 3 June 2026].

Thank You

